

A Knowledge Map of Microbially-Mediated Dolomite Precipitation

Introduction:

Dolomite precipitation is rare in modern Earth-surface conditions, which favor calcite formation (i.e. limestone) (Roberts et al., 2013). Yet, dolomite ($\text{CaMg}(\text{CO}_3)_2$) and dolostone, its lithified equivalent, are common in layers of ancient sedimentary rocks from the Archean and Proterozoic Eons (Chang et al., 2020, Hood and Wallace, 2011). This disparity remains poorly understood and is often referred to as “the dolomite problem” (Arvidon and MacKenzie, 1999). Dolomite contains magnesium ions, which hold water molecules around them, inhibiting nucleation. While laboratory conditions, such as very high temperatures, can overcome the kinetic barrier to dolomite formation, the low temperatures found on Earth’s surfaces (less than 50°C) are unable to do so, abiotically (Chen et al., 2023). Microorganisms, however, appear to be able to overcome the kinetic barriers generated by the presence of magnesium ions, to create conditions that favor dolomite formation (Vasconcelos et al., 1995). Specifically, microbial consortia that are found in microbial mats of hypersaline lakes (Wacey et al., 2007; DiLoreto et al., 2019)). Gaining a better understanding of natural dolomite precipitation can provide us with a better picture of carbonate formation of earth’s past and help improve climate models (Hardie 1996).

Australian geology provides natural laboratories for contemporary dolomite precipitation. In particular, are its hypersaline lakes, which possess unique aqueous geochemistry. Many of these lakes are endorheic, or closed basins that do not empty into the ocean, and are therefore subject to evaporation which increases salinity (Bowen and Benison, 2009). These lakes are home to microbial communities that work together to mediate the precipitation of carbonate minerals, including dolomite, through metabolic pathways, such as sulfate reduction and methanogenesis. Those pathways alter local geochemistry such as pH, alkalinity, and sulfide concentrations, to create an environment that influences which type of carbonate mineral formation is favored (Vasconcelos et al., 1995). In addition, microbes produce molecules on their surfaces that could provide nucleation sites where minerals can begin to precipitate (Bontognali et al., 2010). Contemporary microbially-mediated dolomite formation may be the key to understanding ancient sedimentary rock formation (Vasconcelos et al., 1995; Bontognali et al., 2010). However, so far, while multiple species have been associated with dolomite

precipitation, no specific species has been identified as the key player and the exact mechanism is yet to be determined (Bontognali et al., 2010).

In order to gain a better understanding of the body of data that has been acquired so far about microbially-mediated dolomite precipitation, I created a knowledge map to show the relationships between biological, geochemical, and environmental entities.

Methods: Construction of Cytoscape Knowledge Map

Cytoscape (version 3.10.4) was used to construct the knowledge map (Shannon et al., 2003). Biological, geochemical, and environmental concepts relevant to microbially-mediated dolomite formation were integrated in the map.

Step 1. Data Collection

Key concepts and relationships were identified by reviewing primary scientific literature and review papers, as well as, curated databases from NCBI, UniProt, Gene Ontology, ENVO, and ChEBI. Four major categories were identified as nodes for the network: Biological (i.e. microorganisms, metabolic pathways, enzymes), Geochemical (i.e. minerals, ions, geochemical parameters) and Environmental (i.e. Locations, habitats)

Step 2. Table Construction for Import into Cytoscape

The nodes and edges tables were compiled in Google Sheets, then exported as CSV files for importation into Cytoscape (v 3.10.4)

Final Project Nodes and Edges Tables for Cytoscape

Edges Table. Information was compiled into a Google Sheet that could be exported in CSV format. The sheet was organized into the following columns: Source Node, Edge Type, Target Node, Node Type, Node Category, citation/identifier, and Evidence statement. Edge Type represents the relationship between source and target nodes.

Nodes Table. A table was created in Google Sheets that included the Node ID, Node Type, Node Category, and Identifier. This was exported in CSV format.

Step 3. Network Assembly and Visualisation.

The Nodes and Edges tables were imported into Cytoscape, which generates the network of nodes connected to each other to show relationships. Nodes were color coded by category (Biological- blue, Geochemical- purple, and Environmental- green) Identifiers and citations were added to the Node Details.

Several refinements were made before exporting the final knowledge map. Duplicate nodes were removed, additional edges were added to remove “orphan” nodes. Relationships and key concepts were verified for accuracy and inclusion.

Discussion.

Dolomite precipitation has a significant kinetic barrier, making its occurrence in natural settings less favorable than calcite formation. Multiple species of microorganisms that function together as a consortia can create environmental conditions that facilitate dolomite precipitation by altering geochemical parameters. The dolomite problem remains unsolved because the exact mechanisms and the relative contributions of each member of the microbial consortia have not yet been determined. (Diloreto et al., 2021). While several microbial taxa and functional groups have been associated with the process, and while some specific organisms have been induced to precipitate dolomite in culture, there is no single species that has been identified as the main driver of the process in all settings. (Han et al., 2022)

This project resulted in the creation of a knowledge map (Fig.1) of the known biological, geochemical and environmental factors associated with microbially-mediated dolomite formation in modern environments to highlight remaining knowledge gaps for future study. The creation of the map has highlighted an important factor observed in multiple studies. Microbial extracellular polymeric substances (EPS) appear to play an important role in dolomite formation by providing sites for mineral nucleation, which is the starting point of crystal formation (Bontognali et al., 2010). This is very interesting, because it shows us that the microbes are not only altering geochemical conditions through their metabolic processes, such as sulfate reduction and methanogenesis, but are providing a physical site for crystallization on their surfaces (Vasconcelos et al., 1995; Bontognali et al., 2014). I would like to further explore how exactly these EPS macromolecules could assist with lowering the kinetic barrier put in place due to the

hydration shell surrounding the magnesium ions (Roberts et al., 2013; Fang et al., 2021). What is happening at the chemical level that can displace or replace the water, to free the magnesium? Another question for further research would be to identify specific EPS molecules. An overarching theme of this field is that multiple players, processes and molecules are involved. Teasing it apart will take some time and creative experiments.

There were 2 main challenges that I encountered in creating my network itself. The first was dealing with orphan nodes left floating at the bottom with no connection to the main network. I realized that there had to be multiple edges that would connect one node to another indirectly. It took several attempts before I was able to connect all of the orphan nodes to my map. The next challenge was how to represent hierarchies. There is still much need for editing to improve the hierarchical relationships. An example of this is when trying to describe relationships between species or functional groups with microbial mats and consortia, where there may be specific microbes working together (i.e. sulfate reducing bacteria and methanogenic archaea) (Vasconcelos et al., 1995; Bontognali et al., 2010), and then also deciding whether to separate Bacteria from Archaea or just leave that as one broad category for a node. I also would like to spend more time removing redundant relationships and cleaning it up, for example (“contributes_to” and “facilitates”). Another issue I had difficulty with figuring out how to display was the term microbialite. A microbialite is the result of a Geobiological process, which I did not include as a main category. It is the result of biological processes, yet is also a geological feature and example of carbonate precipitation (Wacey et al., 2007).

For future work, I would like to see temporal changes integrated into the map in addition to the spatial information I've included. There is evidence that changes in carbonate type occur seasonally, as reported in hypersaline flats in Qatar (Diloreto et al., 2021). The layers that I saw in my stained cross sections from maars of western Victoria exhibited alternating dolomite with other carbonate minerals, although we did not investigate the ages of each layer. I would like to understand if that is seasonal, perhaps due to drought vs rainy seasons (the maar lakes are closed basins and subject to evaporation) (Bowen and Benison, 2009, Wacey et al., 2007), or due to changes in temperature (Australia experiences extremely high summer temperatures) or is there something else? Given Australia's experience with extreme weather events due to climate change, it would be interesting to know if microbially-mediated dolomite precipitation in these regions is affected and changes in the future (Hardie 1996, Chen et al., 2023). It would be interesting to incorporate paleoclimate data from the ancient dolostones (Hood and Wallace, 2011), to see if there were connections to modern climate data of that region, and to incorporate C-13 isotope data into the knowledge map to get a clearer picture of biological mediation in

ancient dolostone deposits. Finally, I would like to integrate metagenomic data from these communities to get a clearer picture of what communities are present at each modern-day setting where dolomite precipitation is occurring (DiLoreto et al., 2019; Podell et al., 2014).

Future investigations that can add to and integrate the various pieces of microbial, geochemical and environmental information we have of the dolomite puzzle will be critical to understanding low temperature dolomite precipitation connecting modern occurrence to the rock record and the paleoclimate. It is a fascinating topic and one that continues to pique my interest. I appreciated the opportunity to delve into this topic for this project.

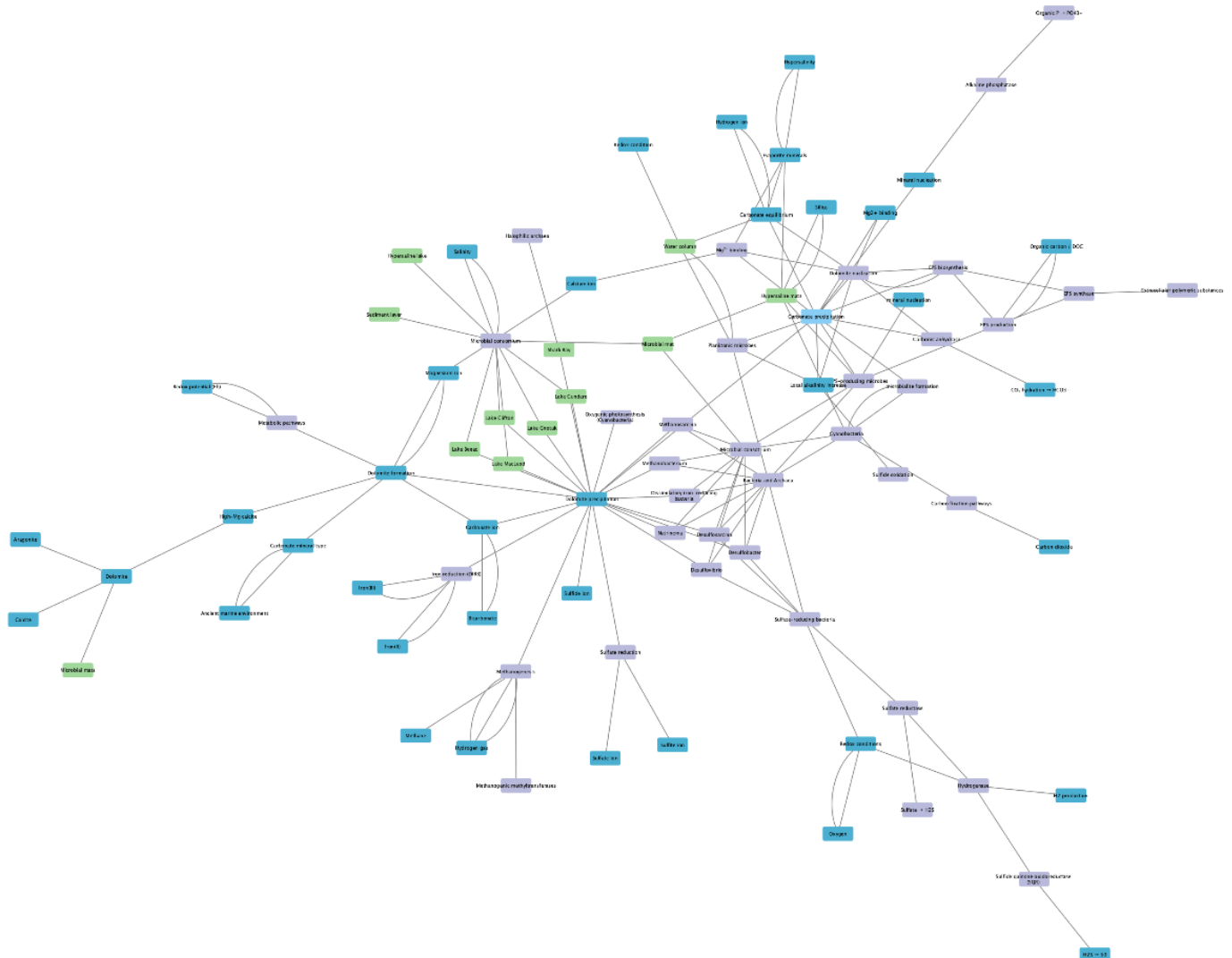


Figure 1. Knowledge map of microbial, geochemical and environmental interactions associated with dolomite precipitation. Nodes represent biological (purple), geochemical (blue), and environmental (green). Edges indicate functional or causal relationships. Interactions were integrated using primary scientific publications and ontology-based data. Visualization was generated using Cytoscape version 3.10.4 (Shannon et al., 2003).

Citations:

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Statement of original work:

"I, Jill E. Lynch, declare that this final project is entirely my own work. I performed all analysis and cited other's work accordingly."